

ONLINE PATHSHALA

CORE JAVA PROGRAMMING

Professional Course Catalog

FROM BEGINNER TO JOB-READY JAVA DEVELOPER

Concept-driven • Practical Coding • Projects • Interview Preparation

Faculty

Ambika Gupta & Shivam Gupta

Duration: 8–10 Weeks | **Mode:** Classroom / Online

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Location: Near Badarpur, New Delhi

COURSE OVERVIEW

Core Java Programming is designed for beginners who want to build a strong foundation in Java and progress toward real-world application development. The course combines clear concepts, live coding, problem solving, projects and interview-oriented practice.

Course Level	Beginner to Intermediate
Course Type	Practical + Hands-on Training
Duration	8–10 Weeks
Prerequisite	Basic Computer Knowledge
Language	Java
Training Focus	Concept Clarity + Coding + Projects

WHY LEARN CORE JAVA?

- Build a strong programming foundation.
- Understand Object-Oriented Programming deeply.
- Develop problem-solving and logical thinking skills.
- Learn collections, exceptions, multithreading and file handling.
- Connect Java applications with databases using JDBC.
- Prepare for Java-based academic, project and interview requirements.

LEARNING APPROACH

Concept → Real-Life Example → Syntax → Live Coding → Output → Debugging → Practice

MODULES 01–05

MODULE 01 — JAVA FUNDAMENTALS

- What is Java?
- Features and applications of Java
- JDK, JRE & JVM
- Java bytecode and platform independence
- Java installation and setup
- First Java program
- Program structure and main() method
- Compilation and execution

MODULE 02 — VARIABLES & DATA TYPES

- Variables and constants
- Primitive and non-primitive data types
- byte, short, int, long
- float, double, char, boolean
- String basics
- Naming conventions
- Widening and narrowing casting

MODULE 03 — OPERATORS

- Arithmetic operators
- Relational operators
- Logical operators
- Assignment operators
- Unary operators
- Practical calculator and billing programs

MODULE 04 — USER INPUT

- Scanner class
- Reading int, double, String and char
- User-input driven programs
- Student information and bill calculator

MODULE 05 — CONTROL FLOW

- if, if-else and nested if
- else-if ladder
- switch, case, break and default
- Even/odd, largest number and grade programs
- Voting eligibility and menu-based programs

MODULES 06–10

MODULE 06 — LOOPS & ITERATION

- for, while and do-while
- Nested loops
- break and continue
- Tables, sums and counting programs
- Factorial, Fibonacci, prime, palindrome and Armstrong programs

MODULE 07 — PATTERN & LOGIC BUILDING

- Star patterns
- Number patterns
- Triangles, pyramids and diamonds
- Logic-building problem sets
- Problem-solving practice

MODULE 08 — ARRAYS

- One-dimensional arrays
- Array input, output and traversal
- Searching and basic operations
- Maximum, minimum, sum and average
- Two-dimensional arrays and matrices
- Matrix addition and multiplication

MODULE 09 — STRINGS

- String creation and methods
- Concatenation and comparison
- equals(), length(), charAt(), substring()
- StringBuilder and StringBuffer
- Reverse, palindrome and character-count programs

MODULE 10 — OBJECT-ORIENTED PROGRAMMING

- Classes and objects
- Methods and instance variables
- Constructors
- Constructor overloading
- this keyword
- Object references

MODULES 11–15

MODULE 11 — INHERITANCE

- Single, multilevel and hierarchical inheritance
- Method inheritance
- super variable, method and constructor

MODULE 12 — POLYMORPHISM

- Method overloading
- Compile-time polymorphism
- Method overriding
- Runtime polymorphism
- Dynamic method dispatch

MODULE 13 — ENCAPSULATION & ABSTRACTION

- Data hiding
- Getters and setters
- Abstract classes and methods
- Interfaces
- Interface vs abstract class

MODULE 14 — STATIC, FINAL & ACCESS MODIFIERS

- static variables, methods and blocks
- final variables, methods and classes
- private, default, protected and public

MODULE 15 — PACKAGES

- Built-in and user-defined packages
- Creating and importing packages
- Access control across packages

MODULES 16–20

MODULE 16 — EXCEPTION HANDLING

- Exception vs Error
- Checked and unchecked exceptions
- try, catch and finally
- Multiple and nested catch
- throw, throws and custom exceptions

MODULE 17 — WRAPPER CLASSES

- Primitive vs wrapper classes
- Integer, Double, Character, Boolean
- Autoboxing and unboxing
- Parsing methods

MODULE 18 — OBJECT CLASS

- toString()
- equals()
- hashCode()
- Reference vs content comparison

MODULE 19 — COLLECTION FRAMEWORK

- Collection architecture
- List: ArrayList, LinkedList, Vector, Stack
- Set: HashSet, LinkedHashSet, TreeSet
- Map: HashMap, LinkedHashMap, TreeMap
- Add, remove, search, update and iteration

MODULE 20 — ITERATOR & GENERICS

- Iterator interface
- hasNext() and next()
- Generic classes and methods
- Type safety and wildcards

MODULES 21–26

MODULE 21 — COMPARABLE & COMPARATOR

- Sorting objects
- Comparable and compareTo()
- Comparator and compare()
- Custom sorting

MODULE 22 — JAVA 8 FEATURES

- Lambda expressions
- Functional interfaces
- Predicate, Consumer, Supplier and Function
- Stream API
- filter, map, sorted, forEach, collect and reduce

MODULE 23 — MULTITHREADING

- Process vs thread
- Creating threads
- Thread class and Runnable
- Thread life cycle
- start(), run(), sleep(), join()
- Synchronization and race conditions

MODULE 24 — FILE HANDLING

- File class
- Reading and writing files
- FileReader and FileWriter
- BufferedReader and BufferedWriter
- try-with-resources

MODULE 25 — SERIALIZATION

- Serialization and deserialization
- Serializable interface
- transient keyword

MODULE 26 — JDBC

- JDBC architecture and drivers
- Database connection
- Statement, PreparedStatement and ResultSet
- CRUD operations
- Java + MySQL integration

PROJECTS & PRACTICAL TRAINING

Students progressively apply concepts through mini-projects and a final database-backed application.

01 — Calculator	Variables, operators, Scanner, switch
02 — Student Result System	Input, conditions, operators and methods
03 — Banking System	Classes, objects, encapsulation and inheritance
04 — Employee Management System	OOP, collections and exception handling
05 — Student Management System	Collections, file handling and OOP
FINAL PROJECT — Java + MySQL Application	OOP, exceptions, collections, JDBC, CRUD and database integration

LEARNING OUTCOMES

- Write Java programs independently from fundamentals to intermediate level.
- Apply OOP concepts to real-world problems.
- Use arrays, strings and collections effectively.
- Handle exceptions and work with files.
- Understand Java 8 functional programming features.
- Build multithreaded programs at a foundational level.
- Connect Java applications with databases using JDBC.
- Develop a complete Java + MySQL project.

START YOUR JAVA JOURNEY

Build Strong Programming Fundamentals. Build Your Java Career.

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COURSE HIGHLIGHTS

- ✓ Beginner Friendly
- ✓ Practical Coding
- ✓ Logic Building
- ✓ OOP Mastery
- ✓ Interview Preparation
- ✓ Hands-on Projects
- ✓ JDBC & Database Integration
- ✓ Final Project

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